Table 1: Efficiency on WearGait asynchronous input (mean±std over 10 repeats on the same GPU, architecture, batch size, seed, and fold).

Method	Time (s/epoch)	Time overhead	Peak memory (MiB)	Memory overhead
TRIP w/o MOO	0.283 ± 0.010	–	20.8 ± 0.0	–
TRIP	0.607 ± 0.028	114.9%	23.4 ± 0.0	12.7%

Available modalities	TRIP latency (ms/window)	Throughput (windows/s)
1	0.581 ± 0.005	1720.2
2	1.116 ± 0.006	895.8
3	1.622 ± 0.014	616.4

Note. Each training repeat uses one untimed warm-up epoch followed by one timed epoch. Peak memory is measured with `torch.cuda.max_memory_allocated`. Overhead is (TRIP − w/o MOO)/w/o MOO × 100%. Inference uses batch size 1; inactive branches are not executed. The 1- and 2-modality rows average all three corresponding modality combinations.